

30 Days SQL Micro Course

Practise Sql Queries 3

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Query all columns for all American cities in the **CITY** table with populations larger than 100000.

The **CountryCode** for America is USA.

The **CITY** table is described as follows:

CITY



```
1
2 SELECT *
3 FROM CITY
4 WHERE COUNTRYCODE='USA'
5 AND POPULATION > 100000;
```

Line: 2 Col: 1

Upload Code as File Run Code Submit Code

SELECT *

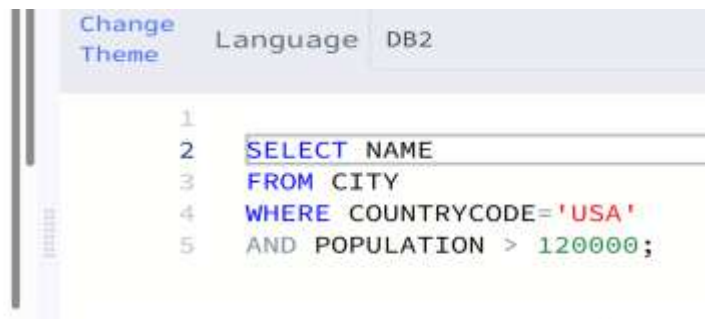
FROM CITY

WHERE COUNTRYCODE='USA'

AND POPULATION > 100000;

Query the **NAME** field for all American cities in the **CITY** table with populations larger than 120000. The CountryCode for America is USA.

The **CITY** table is described as follows:



```
1
2 SELECT NAME
3 FROM CITY
4 WHERE COUNTRYCODE='USA'
5 AND POPULATION > 120000;
```

SELECT NAME

FROM CITY

WHERE COUNTRYCODE='USA'

AND POPULATION > 120000;

Query all columns (attributes) for every row in the **CITY** table.

The **CITY** table is described as follows:



```
1
2 SELECT *
3 FROM CITY;
```

SELECT *

FROM CITY;

Query all columns for a city in **CITY** with the ID 1661.

The **CITY** table is described as follows:



The screenshot shows a SQL editor with a top bar containing 'Change Theme', 'Language', and 'DB2'. The query is displayed in a text area with line numbers 1, 2, and 3. The query text is: `SELECT *` on line 1, `FROM CITY` on line 2, and `WHERE ID = 1661;` on line 3. The text is color-coded: 'SELECT' is blue, '*' is black, 'FROM' is blue, 'CITY' is black, 'WHERE' is blue, 'ID' is black, '1661' is green, and ';' is black.

```
1 SELECT *
2 FROM CITY
3 WHERE ID = 1661;
```

SELECT *

FROM CITY

WHERE ID = 1661;

Query all attributes of every Japanese city in the **CITY** table.

The **COUNTRYCODE** for Japan is JPN.

The **CITY** table is described as follows:



The screenshot shows a SQL editor with a top bar containing 'Change Theme', 'Language', and 'DB2'. The query is displayed in a text area with line numbers 1, 2, and 3. The query text is: `SELECT *` on line 1, `FROM CITY` on line 2, and `WHERE COUNTRYCODE='JPN';` on line 3. The text is color-coded: 'SELECT' is blue, '*' is black, 'FROM' is blue, 'CITY' is black, 'WHERE' is blue, 'COUNTRYCODE' is black, ''JPN'' is red, and ';' is black.

```
1 SELECT *
2 FROM CITY
3 WHERE COUNTRYCODE='JPN';
```

SELECT *

FROM CITY

WHERE COUNTRYCODE='JPN';

Query the names of all the Japanese cities in the **CITY** table.

The **COUNTRYCODE** for Japan is JPN.

The **CITY** table is described as follows:



The screenshot shows a SQL editor with a top bar containing 'Change Theme', 'Language', and 'DB2'. The query is displayed in a text area with line numbers 1, 2, 3, and 4. The query text is: `SELECT NAME` on line 1, `FROM CITY` on line 2, and `WHERE COUNTRYCODE='JPN';` on line 3. The text is color-coded: 'SELECT' is blue, 'NAME' is black, 'FROM' is blue, 'CITY' is black, 'WHERE' is blue, 'COUNTRYCODE' is black, ''JPN'' is red, and ';' is black.

```
1 SELECT NAME
2 FROM CITY
3 WHERE COUNTRYCODE='JPN';
4
```

SELECT NAME

FROM CITY

WHERE COUNTRYCODE='JPN';

Query a list of **CITY** and **STATE** from the **STATION** table.

The **STATION** table is described as follows:

```
Change Theme Language DB2
1 SELECT CITY, STATE
2 FROM STATION;
```

SELECT CITY, STATE

FROM STATION;

Query a list of **CITY** names from **STATION** for cities that have an even **ID** number. Print the results in any order, but exclude duplicates from the answer.

The **STATION** table is described as follows:

```
Change Theme Language DB2
1 SELECT DISTINCT CITY
2 FROM STATION
3 WHERE MOD(ID, 2) = 0;
```

SELECT DISTINCT CITY

FROM STATION

WHERE MOD(ID, 2) = 0;

Find the difference between the total number of **CITY** entries in the table and the number of distinct **CITY** entries in the table.

The **STATION** table is described as follows:

```
Change Theme Language DB2
1 SELECT COUNT(CITY) - COUNT(DISTINCT CITY)
2 FROM STATION;
```

SELECT COUNT(CITY)-COUNT(DISTINCT CITY)

FROM STATION;

Query the two cities in **STATION** with the shortest and longest **CITY** names, as well as their respective lengths (i.e.: number of characters in the name). If there is more than one smallest or largest city, choose the one that comes first when ordered alphabetically.

The **STATION** table is described as follows:

STATION	
Field	Type

```
Change Theme Language DB2
1 (SELECT CITY, LENGTH(CITY)
2 FROM STATION
3 ORDER BY LENGTH(CITY), CITY
4 LIMIT 1)
5
6 UNION
7
8 (SELECT CITY, LENGTH(CITY)
9 FROM STATION
10 ORDER BY LENGTH(CITY) DESC, CITY
11 LIMIT 1);
```

```
(SELECT CITY, LENGTH(CITY)
FROM STATION
ORDER BY LENGTH(CITY), CITY
LIMIT 1)
```

UNION

```
(SELECT CITY, LENGTH(CITY)
FROM STATION
ORDER BY LENGTH(CITY) DESC, CITY
LIMIT 1);
```

Short Explanation

First SELECT

```
ORDER BY LENGTH(CITY), CITY
LIMIT 1
```

- Sorts cities by smallest length first.
- If lengths are equal, sorts alphabetically.
- LIMIT 1 picks the shortest city.

Second SELECT

```
ORDER BY LENGTH(CITY) DESC, CITY
LIMIT 1
```

- DESC sorts by largest length first.
- If lengths are equal, sorts alphabetically.
- LIMIT 1 picks the longest city.

UNION

UNION

- Combines both results into one output.

Example Output

Agra 4

Chennai 8

- Agra 4 → shortest city name.
- Chennai 8 → longest city name.

Query the list of CITY names starting with vowels (i.e., a, e, i, o, or u) from **STATION**.

Your result cannot contain duplicates.

Input Format

The **STATION** table is described as follows:

STATION



SELECT DISTINCT CITY

FROM STATION

WHERE CITY LIKE 'A%'

OR CITY LIKE 'E%'

OR CITY LIKE 'I%'

OR CITY LIKE 'O%'

OR CITY LIKE 'U%';

OR

Query

SELECT DISTINCT CITY

FROM STATION

WHERE LEFT(CITY,1) IN ('A','E','I','O','U','a','e','i','o','u');

Explanation

SELECT DISTINCT CITY

- Selects the CITY column.
- DISTINCT removes duplicate city names.

FROM STATION

- Reads data from the STATION table.

LEFT(CITY,1)

- **LEFT(string, n)** returns the first n characters from the left.
- **LEFT(CITY,1)** gets the first letter of the city name.

Example:

CITY LEFT(CITY,1)

Agra A

Delhi D

Indore I

IN (...)

Checks whether the first letter is one of:

('A','E','I','O','U','a','e','i','o','u')

That is, any vowel (uppercase or lowercase).

OR

Query

SELECT DISTINCT CITY

FROM STATION

WHERE CITY REGEXP '^[AEIOUaeiou]';

Short Explanation

- **DISTINCT** → removes duplicate city names.

WHERE CITY REGEXP '^[AEIOUaeiou]'

Explanation

REGEXP

- **Used to match a pattern using a regular expression.**

'^'

- **Means start of the string.**
- **Checks only the first character of the city name.**

'[AEIOUaeiou]'

- **Means the first character can be:**

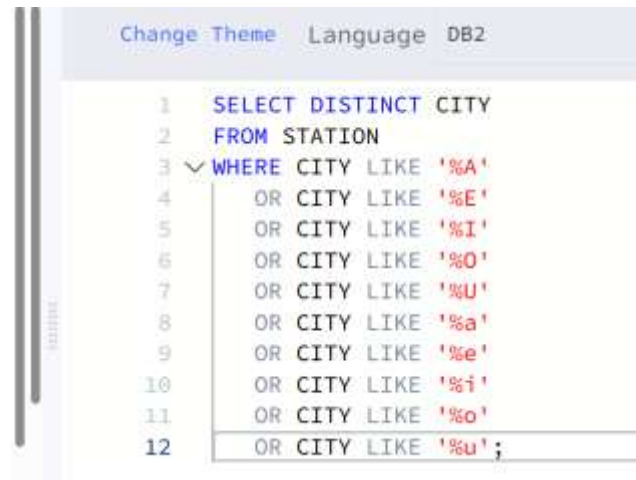
- A, E, I, O, U (uppercase)
- a, e, i, o, u (lowercase)

Query the list of CITY names ending with vowels (a, e, i, o, u) from **STATION**. Your result cannot contain duplicates.

Input Format

The **STATION** table is described as follows:

STATION	
Field	Type
ID	NUMBER



1. Using LIKE

```

SELECT DISTINCT CITY
FROM STATION
WHERE CITY LIKE '%A'
      OR CITY LIKE '%E'
      OR CITY LIKE '%I'
      OR CITY LIKE '%O'
      OR CITY LIKE '%U'
      OR CITY LIKE '%a'
      OR CITY LIKE '%e'
      OR CITY LIKE '%i'
      OR CITY LIKE '%o'
      OR CITY LIKE '%u';

```

%A means the city ends with A (% = any characters before it).

2. Using REGEXP

```

SELECT DISTINCT CITY
FROM STATION
WHERE CITY REGEXP '[AEIOUaeiou]$';

```

Explanation:

- [AEIOUaeiou] → any vowel
- \$ → end of the string

So it means: CITY ends with a vowel.

3. Using RIGHT()

```
SELECT DISTINCT CITY
FROM STATION
WHERE RIGHT(CITY,1) IN ('A','E','I','O','U',
                        'a','e','i','o','u');
```

Explanation:

- RIGHT(CITY,1) gets the last character of the city name.
 - IN (...) checks if the last character is a vowel.
-

Summary

Method Condition

LIKE CITY LIKE '%A' OR ...

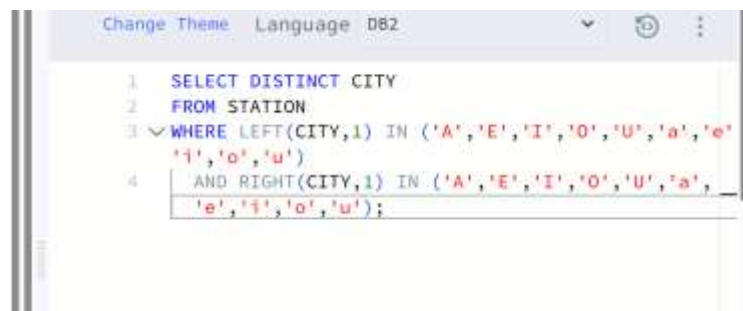
REGEXP CITY REGEXP '[AEIOUaeiou]\$'

RIGHT RIGHT(CITY,1) IN ('A','E','I','O','U','a','e','i','o','u')

Query the list of CITY names from **STATION** which have vowels (i.e., a, e, i, o, and u) as both their first and last characters. Your result cannot contain duplicates.

Input Format

The **STATION** table is described as follows:



1. Using LIKE

```
SELECT DISTINCT CITY
FROM STATION
WHERE (CITY LIKE 'A%' OR CITY LIKE 'E%' OR CITY LIKE 'I%' OR CITY LIKE 'O%' OR
CITY LIKE 'U%')
AND (CITY LIKE '%A' OR CITY LIKE '%E' OR CITY LIKE '%I' OR CITY LIKE '%O' OR CITY
LIKE '%U');
```

Meaning:

- First letter is a vowel.

- Last letter is also a vowel.

2. Using REGEXP

```
SELECT DISTINCT CITY
FROM STATION
WHERE CITY REGEXP '^[AEIOUaeiou].*[AEIOUaeiou]$';
```

Explanation

- ^ → start of string
- [AEIOUaeiou] → first character is a vowel
- .* → any characters in between
- [AEIOUaeiou] → last character is a vowel
- \$ → end of string

3. Using LEFT() and RIGHT()

```
SELECT DISTINCT CITY
FROM STATION
WHERE LEFT(CITY,1) IN ('A','E','I','O','U','a','e','i','o','u')
AND RIGHT(CITY,1) IN ('A','E','I','O','U','a','e','i','o','u');
```

Explanation

- LEFT(CITY,1) → gets the first letter.
- RIGHT(CITY,1) → gets the last letter.
- Both must be vowels.

Query the list of CITY names from **STATION** that do not start with vowels. Your result cannot contain duplicates.

Input Format

The **STATION** table is described as follows:

	Change Theme	Language	DB2
1	SELECT	DISTINCT	CITY
2	FROM	STATION	
3	WHERE	CITY NOT LIKE	'A%'
4		AND CITY NOT LIKE	'E%'
5		AND CITY NOT LIKE	'I%'
6		AND CITY NOT LIKE	'O%'
7		AND CITY NOT LIKE	'U%';

1. Using LIKE

```
SELECT DISTINCT CITY
FROM STATION
```

```
WHERE CITY NOT LIKE 'A%'
AND CITY NOT LIKE 'E%'
AND CITY NOT LIKE 'I%'
AND CITY NOT LIKE 'O%'
AND CITY NOT LIKE 'U%';
```

Meaning: Select cities whose first letter is not A, E, I, O, or U.

2. Using REGEXP

```
SELECT DISTINCT CITY
FROM STATION
WHERE CITY NOT REGEXP '^[AEIOUaeiou]';
```

Explanation:

- ^ → start of string
 - [AEIOUaeiou] → any vowel
 - NOT REGEXP → city does not start with a vowel
-

3. Using LEFT()

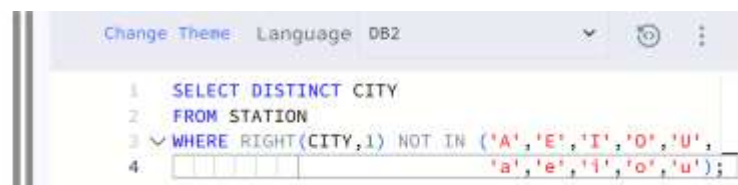
```
SELECT DISTINCT CITY
FROM STATION
WHERE LEFT(CITY,1) NOT IN ('A','E','I','O','U',
                           'a','e','i','o','u');
```

Explanation:

- LEFT(CITY,1) gets the first letter.
- NOT IN checks that the first letter is not a vowel.

Query the list of CITY names from **STATION** that do not end with vowels. Your result cannot contain duplicates.

Input Format



1. Using LIKE

```
SELECT DISTINCT CITY
FROM STATION
WHERE CITY NOT LIKE '%A'
AND CITY NOT LIKE '%E'
```

AND CITY NOT LIKE '%I'
AND CITY NOT LIKE '%O'
AND CITY NOT LIKE '%U';

Meaning: Select cities whose last letter is not A, E, I, O, or U.

2. Using REGEXP

SELECT DISTINCT CITY
FROM STATION
WHERE CITY NOT REGEXP '[AEIOUaeiou]\$';

Explanation:

- **[AEIOUaeiou] → any vowel**
 - **\$ → end of string**
 - **NOT REGEXP → city does not end with a vowel**
-

3. Using RIGHT()

SELECT DISTINCT CITY
FROM STATION
WHERE RIGHT(CITY,1) NOT IN ('A','E','I','O','U',
'a','e','i','o','u');

Explanation:

- **RIGHT(CITY,1) gets the last letter.**
- **NOT IN checks that the last letter is not a vowel.**